

Exam 4 Topics / Sections

- Fluid Pressure
- Area MOI
 - Integration
 - Composite Area
- POI (I_{xy})
 - Integration
 - Composite Area
- Mohr's Circle
 - Usually "simple" probs

Exam 4 Crib Sheet Review

①

Page 1: Fluid Pressure: $P = \gamma z = \rho g z$

Doesn't give fresh water γ 's:

$$\gamma = 62.4 \text{ lb/ft}^3$$

$$\gamma = 9810 \text{ N/m}^3$$

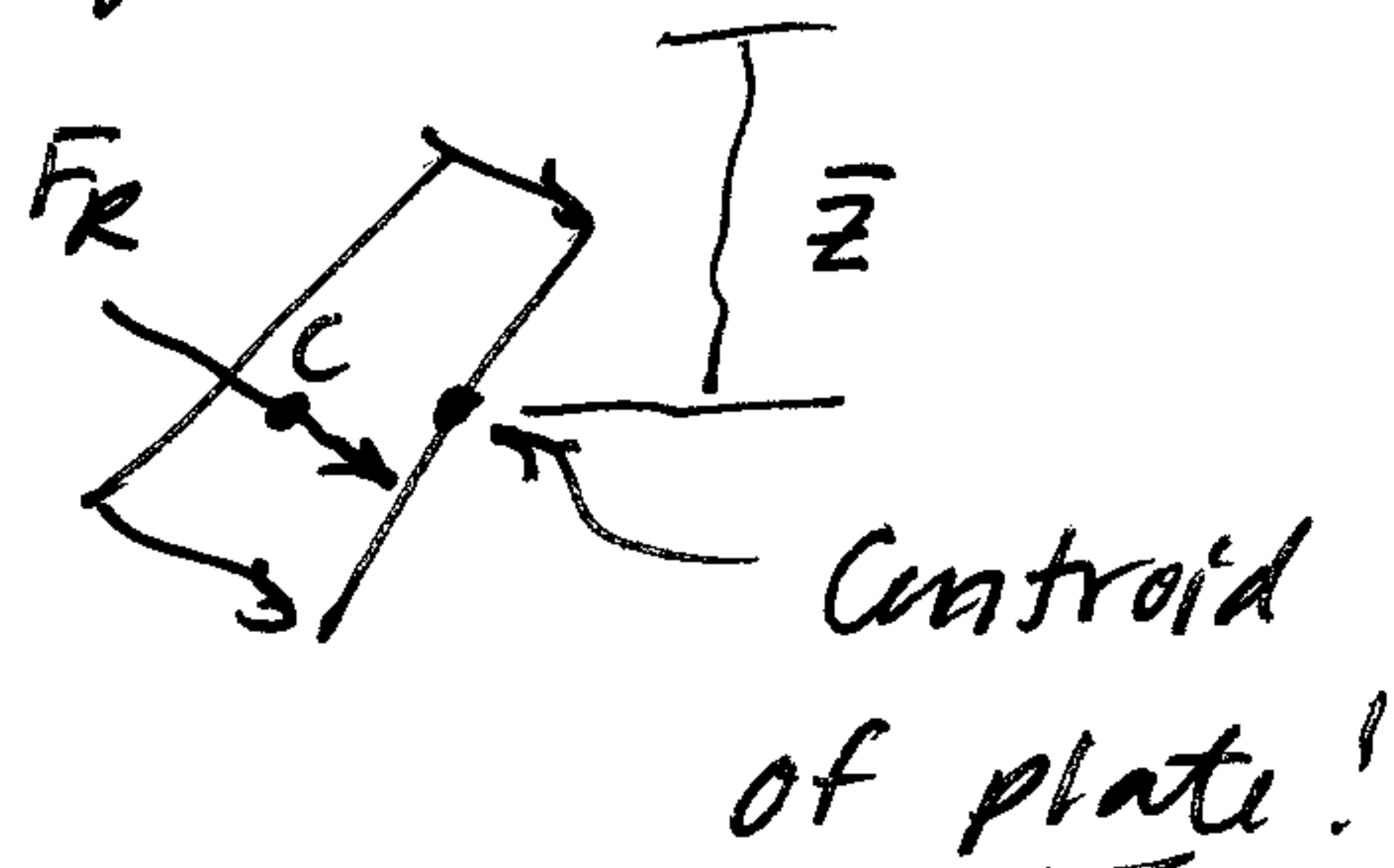
$$= 1000 \text{ kg/m}^3 \cdot 9.81$$

Given: $F_R = \gamma \cdot \bar{z} \cdot A$

But picture doesn't show $\bar{z}$... What is $\bar{z}$?

Shows F_R at centroid $\underline{C}$ of pressure distrib,

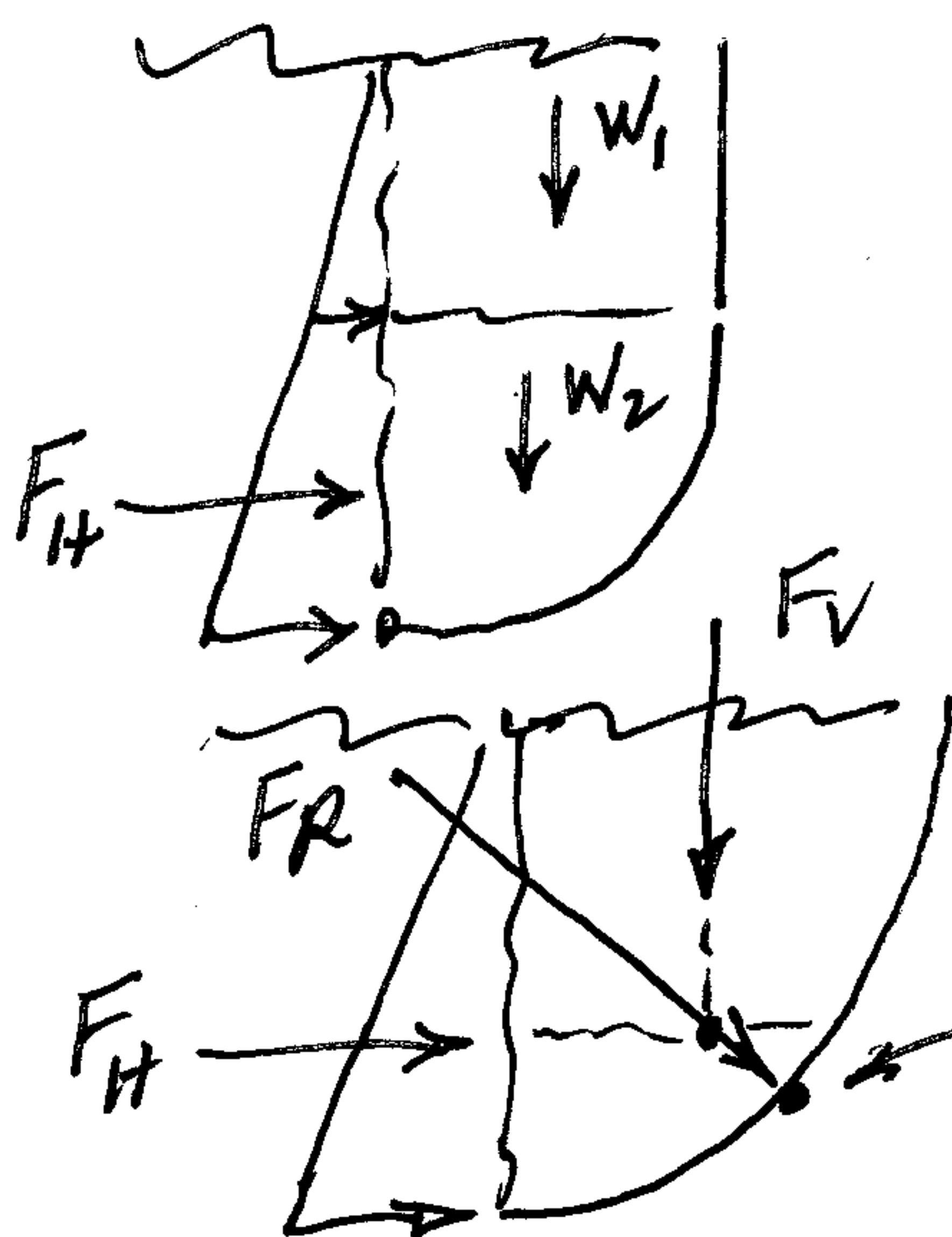
but $\bar{z}$ measured to centroid of plate!



Gives "Composite" Method for curved surfaces:

$$\vec{F}_R = \vec{F}_H + \vec{F}_V$$

• Doesn't say that $F_{AB} = \text{Weight}$



"Center of Pressure" = Point at which $\vec{F}_R$ touches wall.

Crib Sheet Review (cont'd)

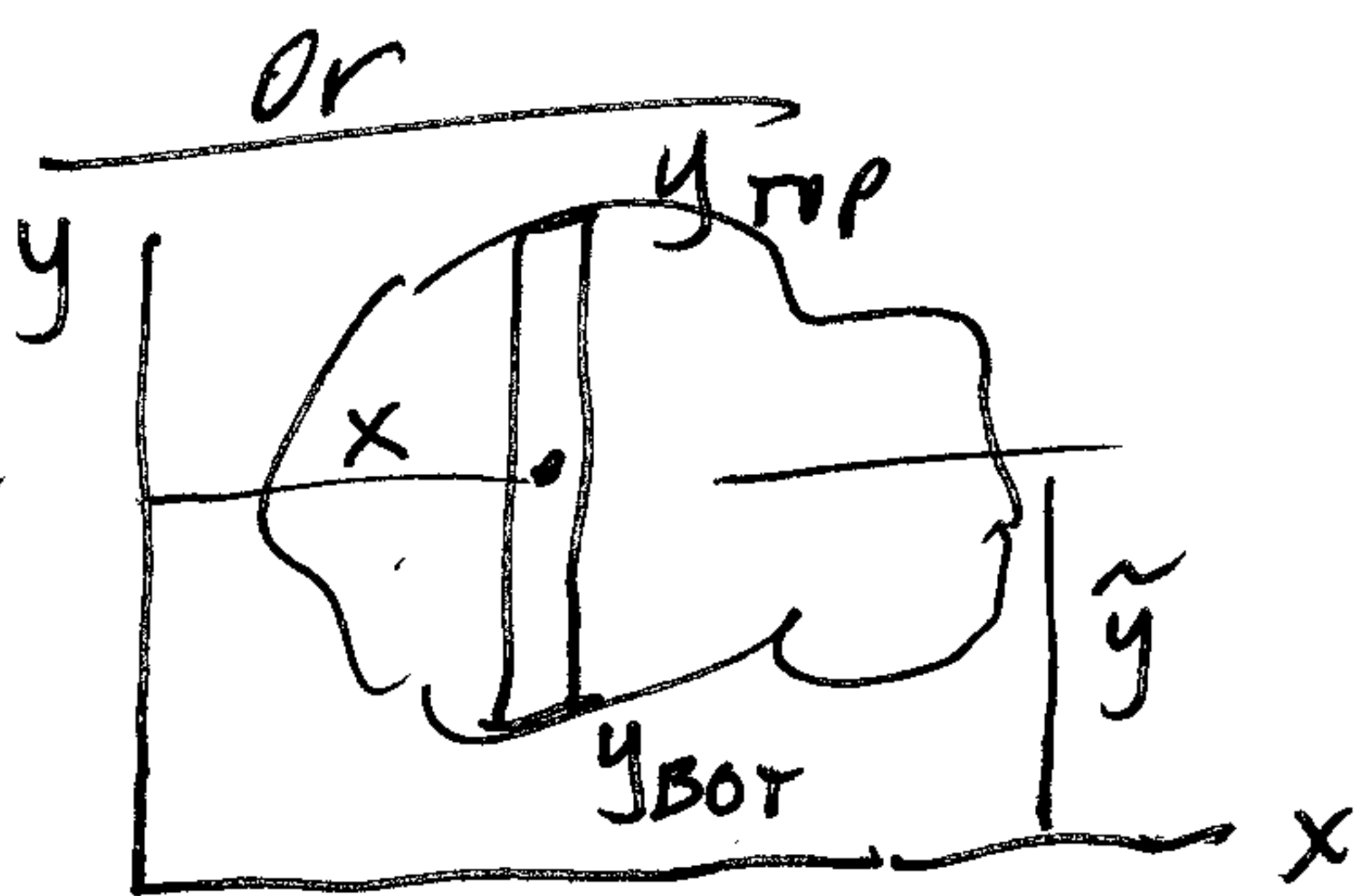
Moment of Inertia

$$I_x = \int_A y^2 dA$$

$$I_y = \int_A x^2 dA$$

Really only true for properly chosen dA :

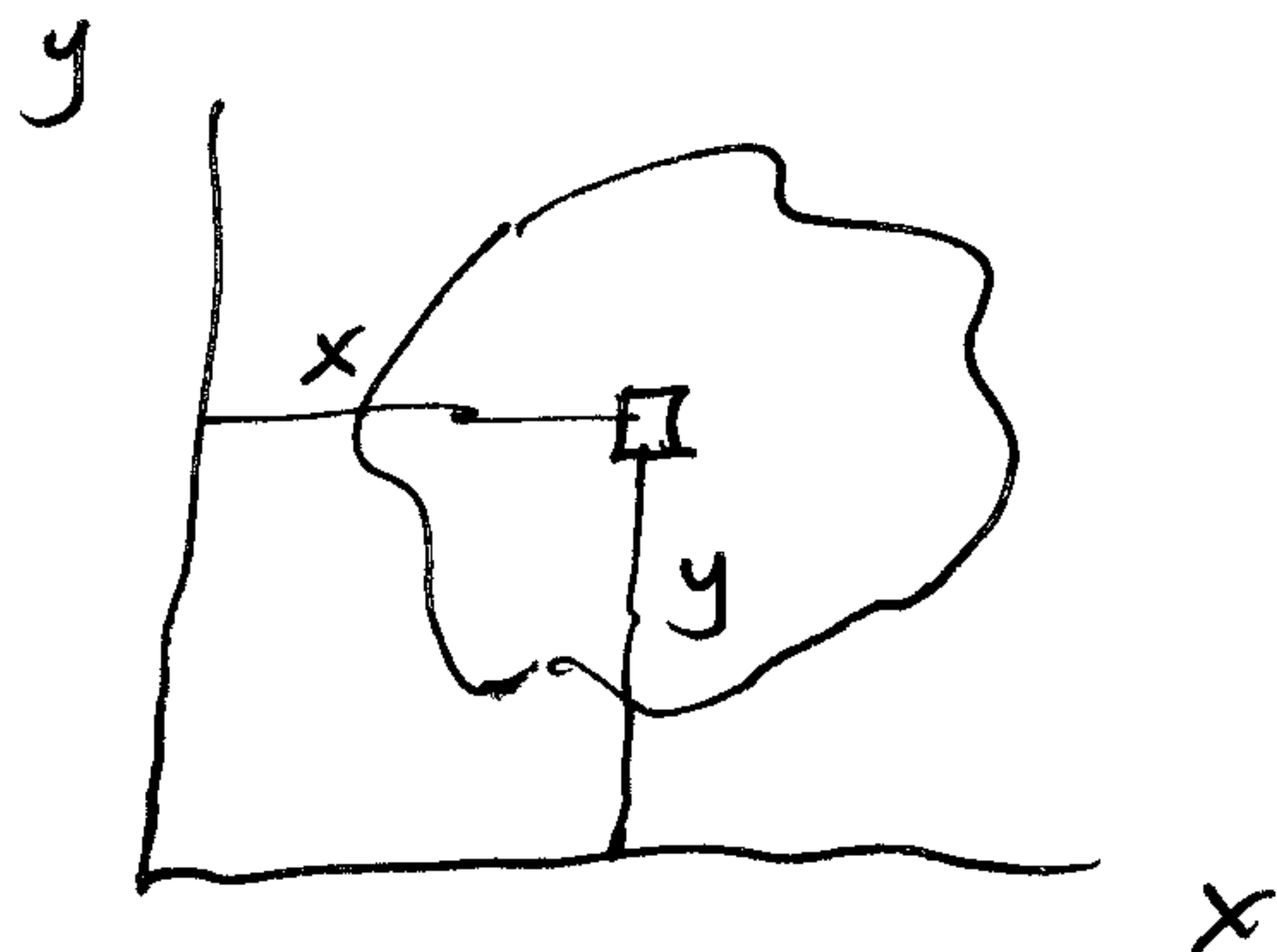
- $dA = dx dy = dy dx$
- dA parallel to x for I_x
- dA parallel to y for I_y



$$\tilde{y} = \left(\frac{y_{\text{TOP}} + y_{\text{BOT}}}{2} \right)$$

$$dA = \left[(y_{\text{TOP}} - y_{\text{BOT}}) \cdot dx \right]$$

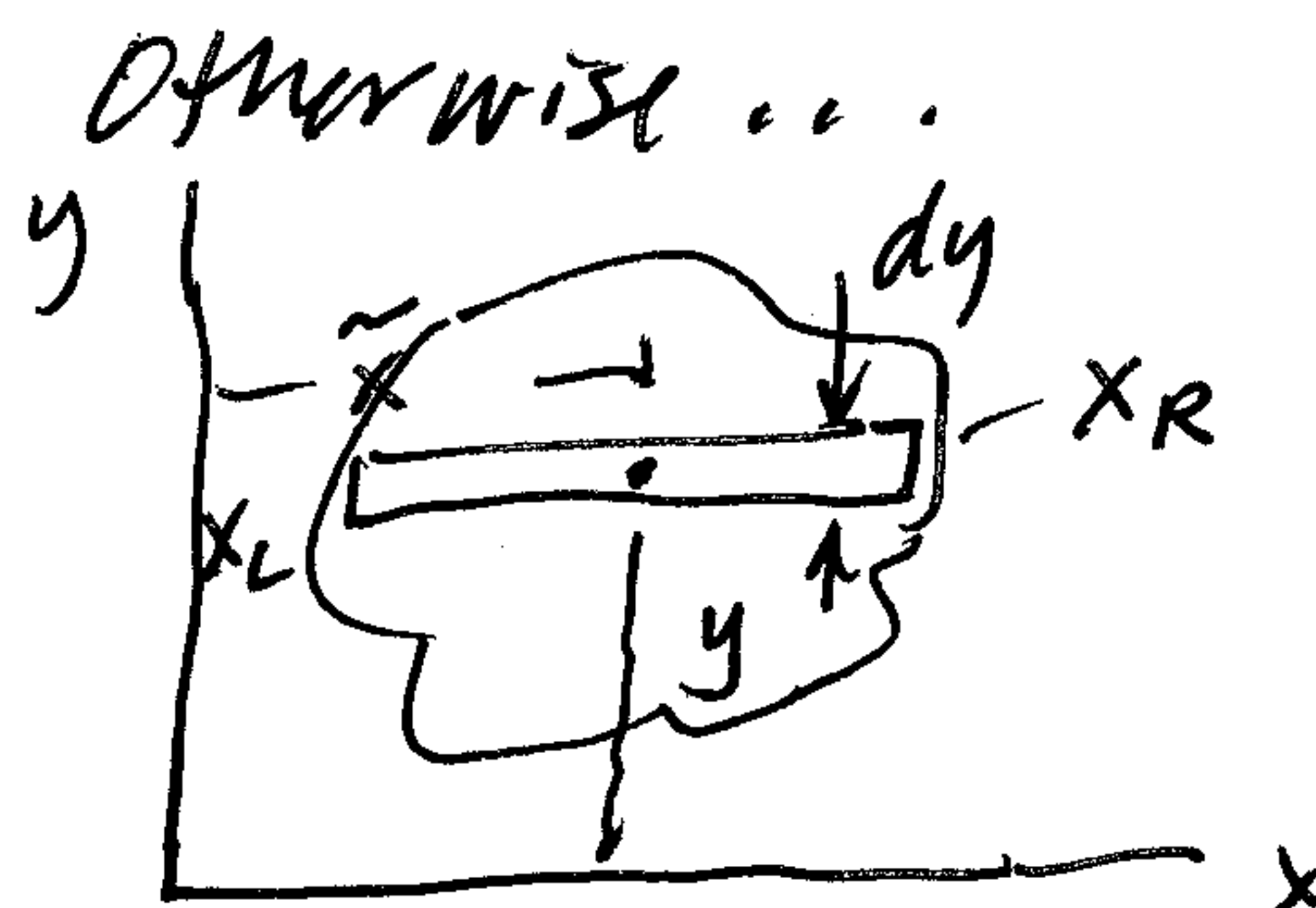
$$I_{xy} = \int x \left(\frac{y_{\text{TOP}} + y_{\text{BOT}}}{2} \right) dA$$



Product of Inertia (POI)

$$I_{xy} = \int_A xy dA$$

↳ only true for double integral and $dA = dx dy$ or $dy dx$



$$\tilde{x} = \left(\frac{x_L + x_R}{2} \right)$$

$$[dA = (x_R - x_L) dy]$$

$$I_{xy} = \int y \left(\frac{x_L + x_R}{2} \right) dA$$

Crib Sheet Review

③

Parallel Axis Theorem (PAT)

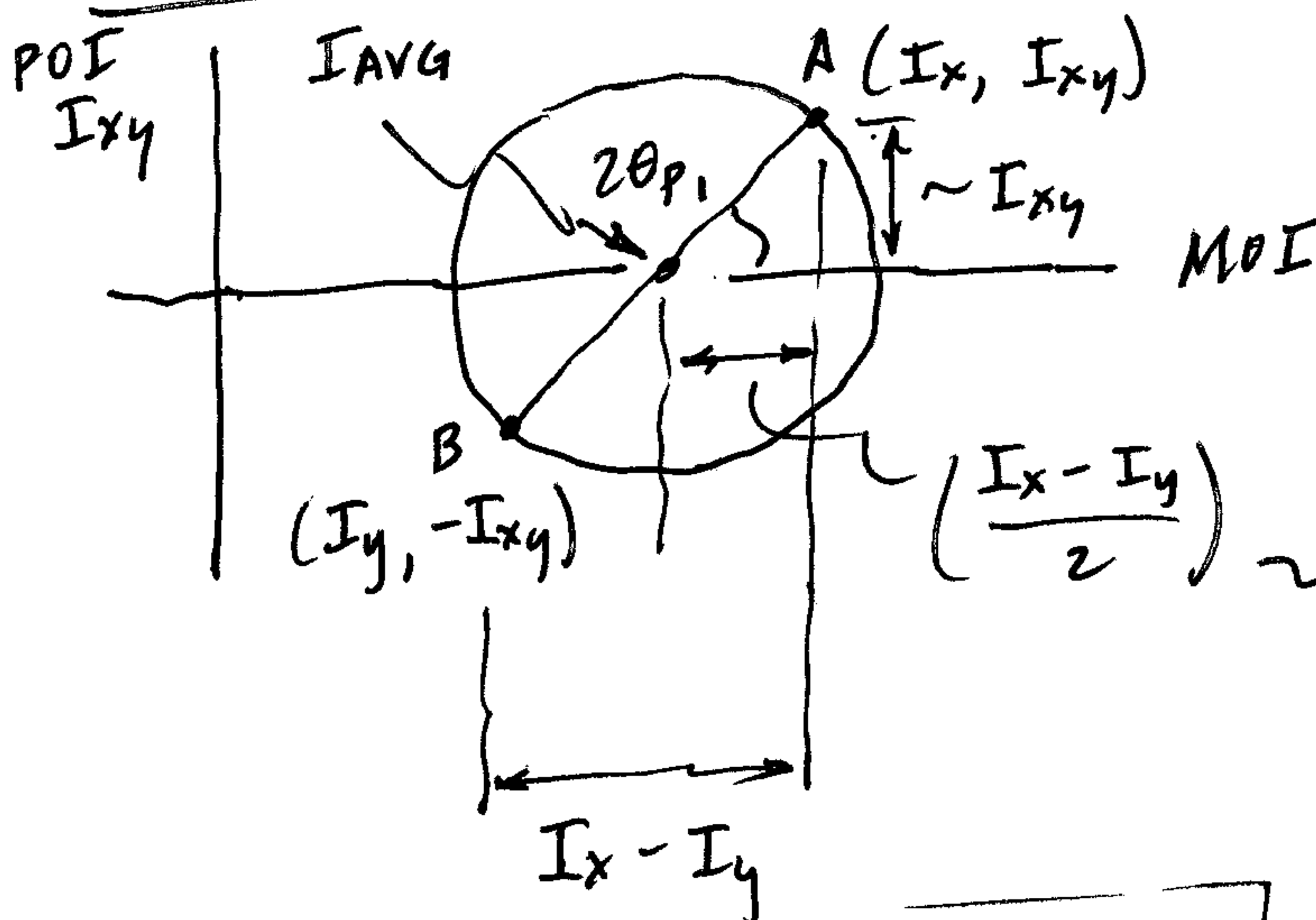
MOI: $I = \bar{I} + Ad^2$

POI: $I_{xy} = \bar{I}_{x'y'} + A dx dy$

Transformation Eqs

Just gives 2θ forms...

Mohr's Circle



Center:

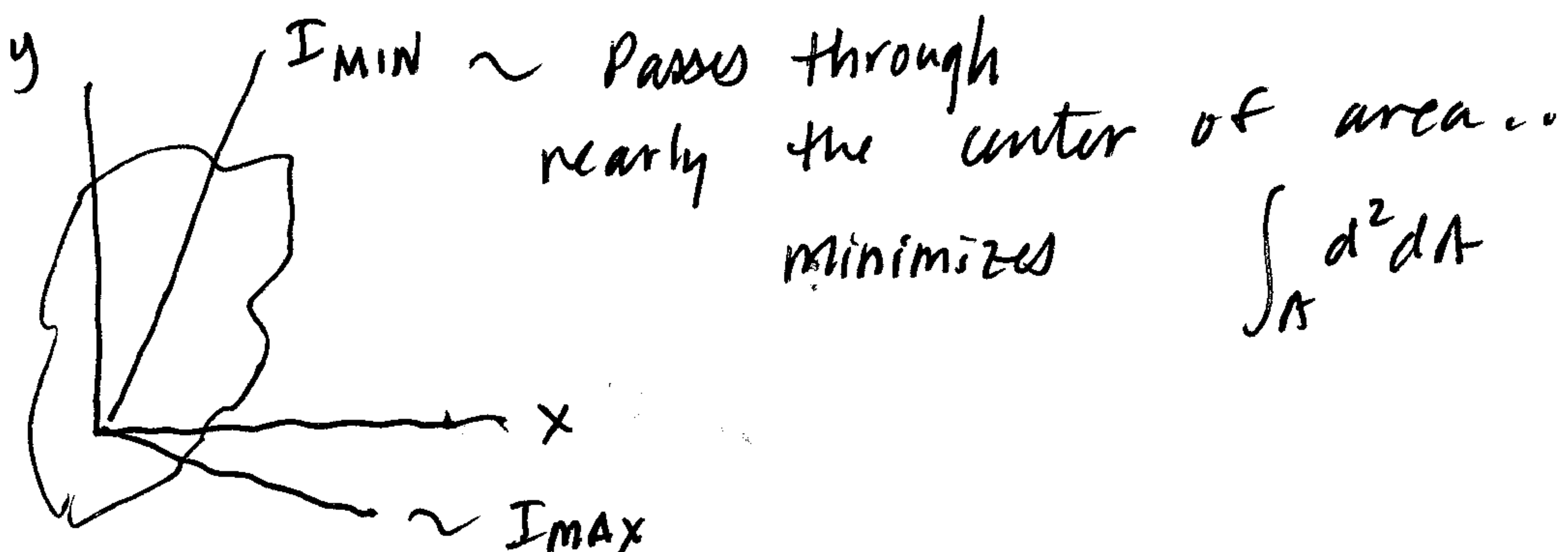
$$I_{AVG} = \frac{I_x + I_y}{2}$$

Also, this is equal to $(I_x - I_{AVG})$

$$R = \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$$

" $(I_x - I_{AVG})$

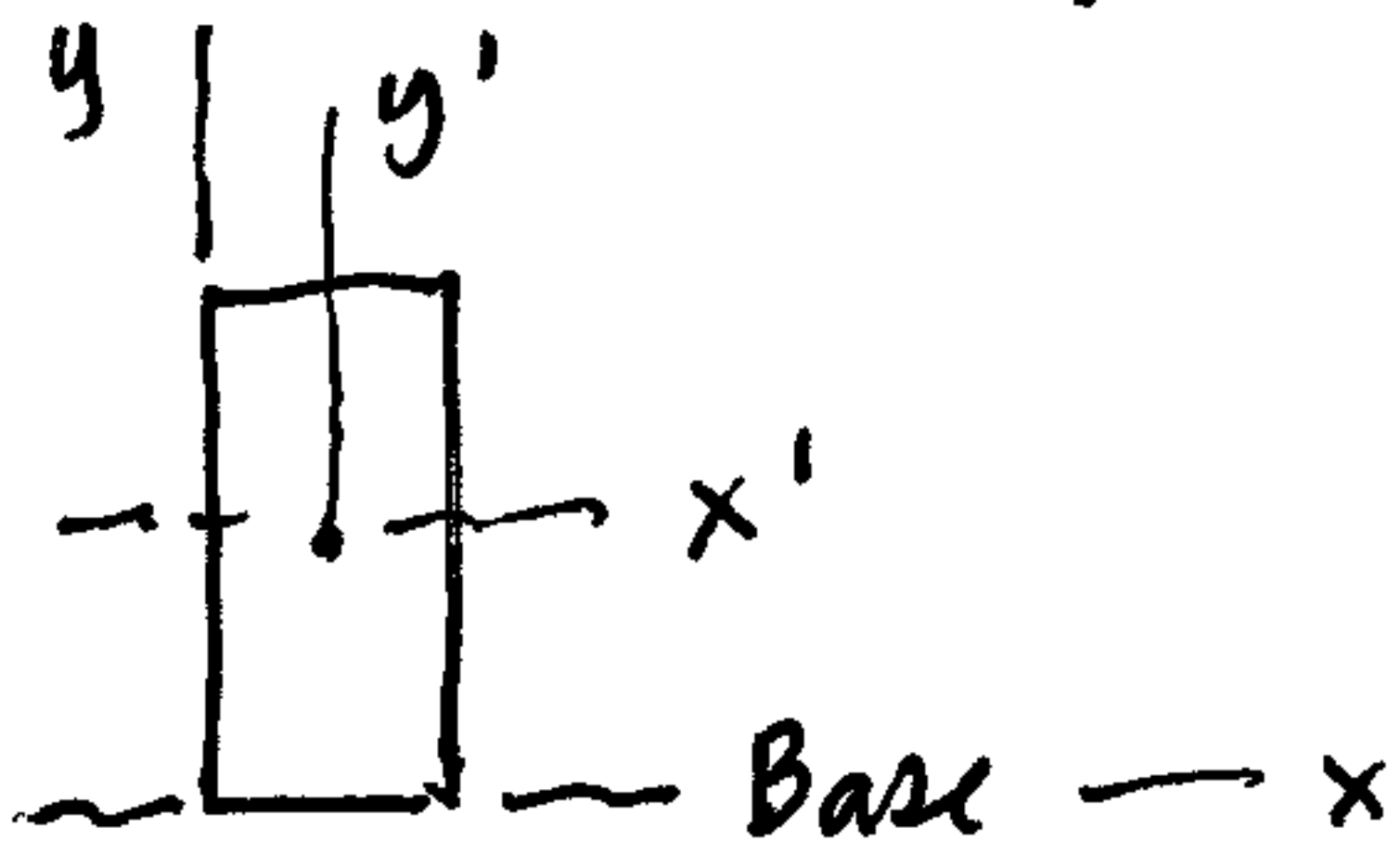
$$\tan 2\theta_p = \frac{-I_{xy}}{\left(\frac{I_x - I_y}{2}\right)}$$



Crib Sheet Review

(p.2)

Gives I_x, I_y, I_{xy} eqns again.



$$\bar{I} = \frac{1}{12} b h^3$$

$$I_{base} = \frac{1}{3} b h^3$$

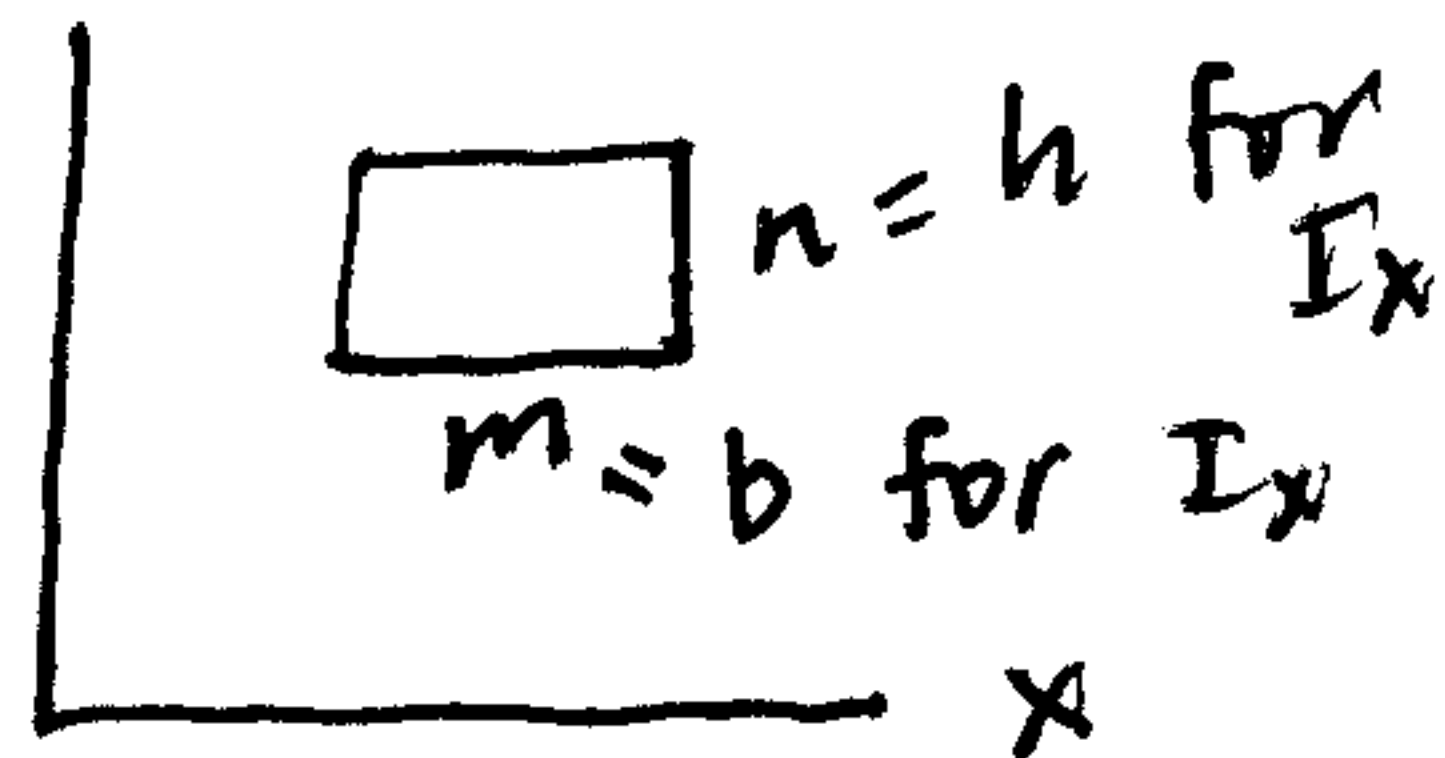
$$\bar{I}_{xy} = 0 \quad \text{Symmetric}$$

I_{origin}



$$I_{xy} = \frac{1}{4} b^2 h^2$$

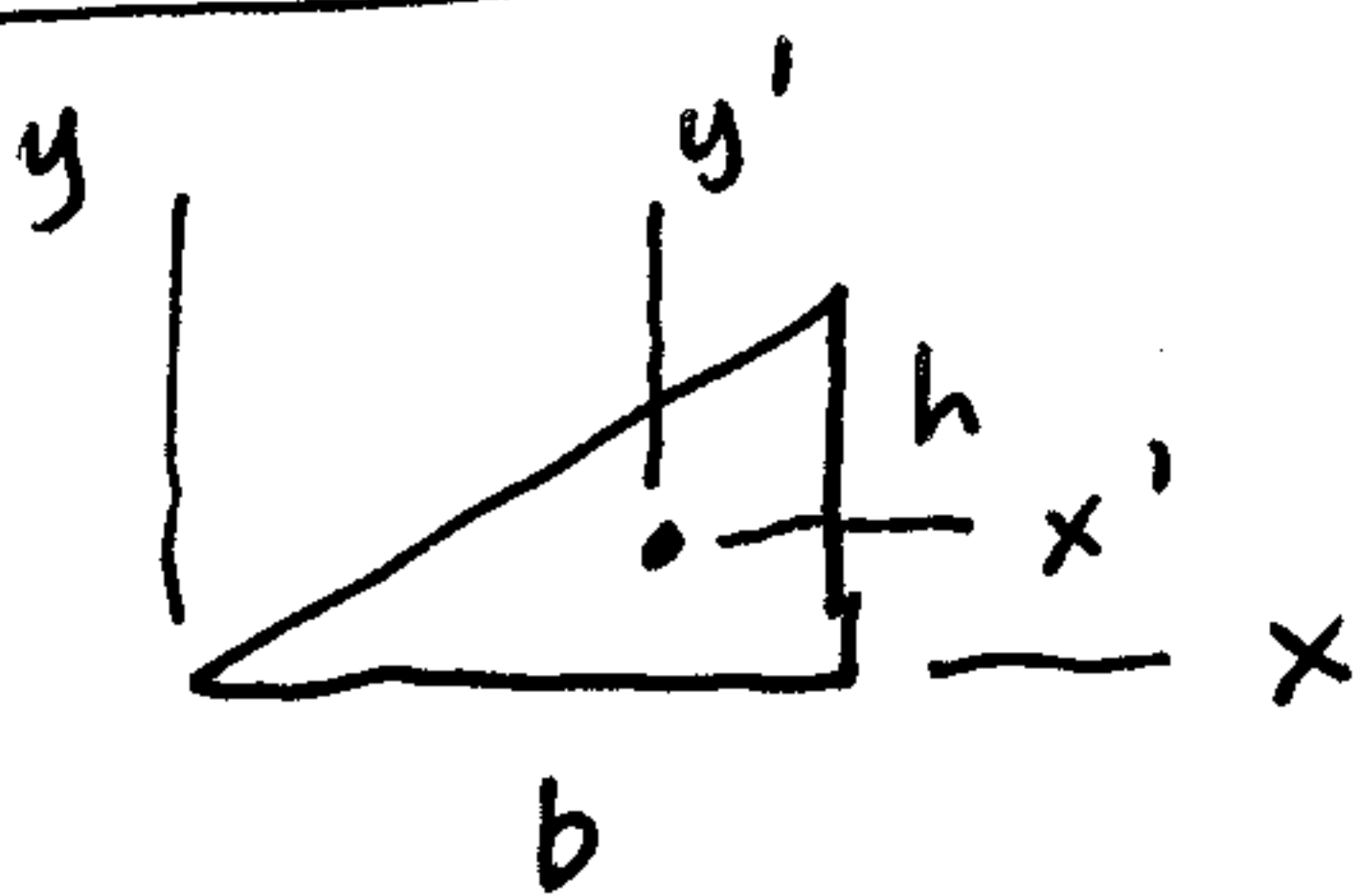
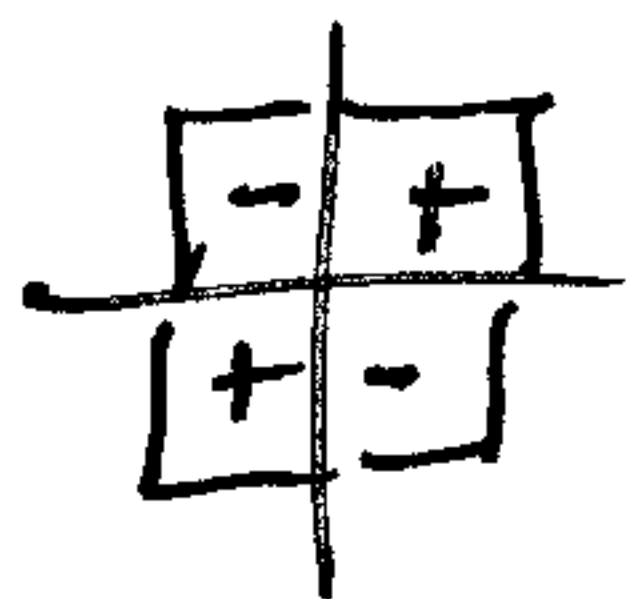
④



$b = \text{dimension parallel to axis}$

$h = \text{dimension } \perp \text{ to axis}$

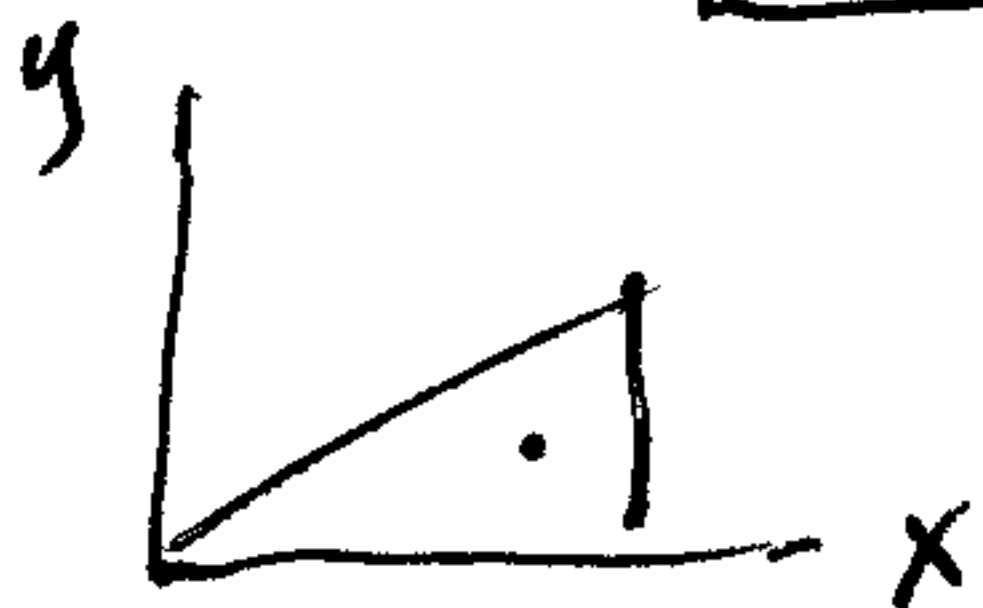
Signs!



$$\bar{I} = \frac{1}{36} b h^3$$

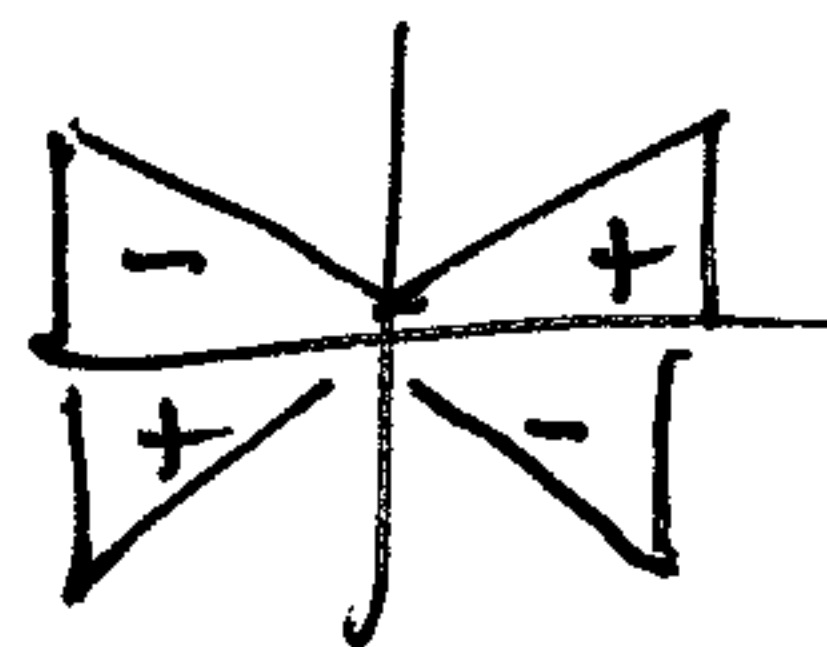
$$I_{base} = \frac{1}{12} b h^3$$

I_{origin}



$$I_{xy} = \frac{1}{8} b^2 h^2$$

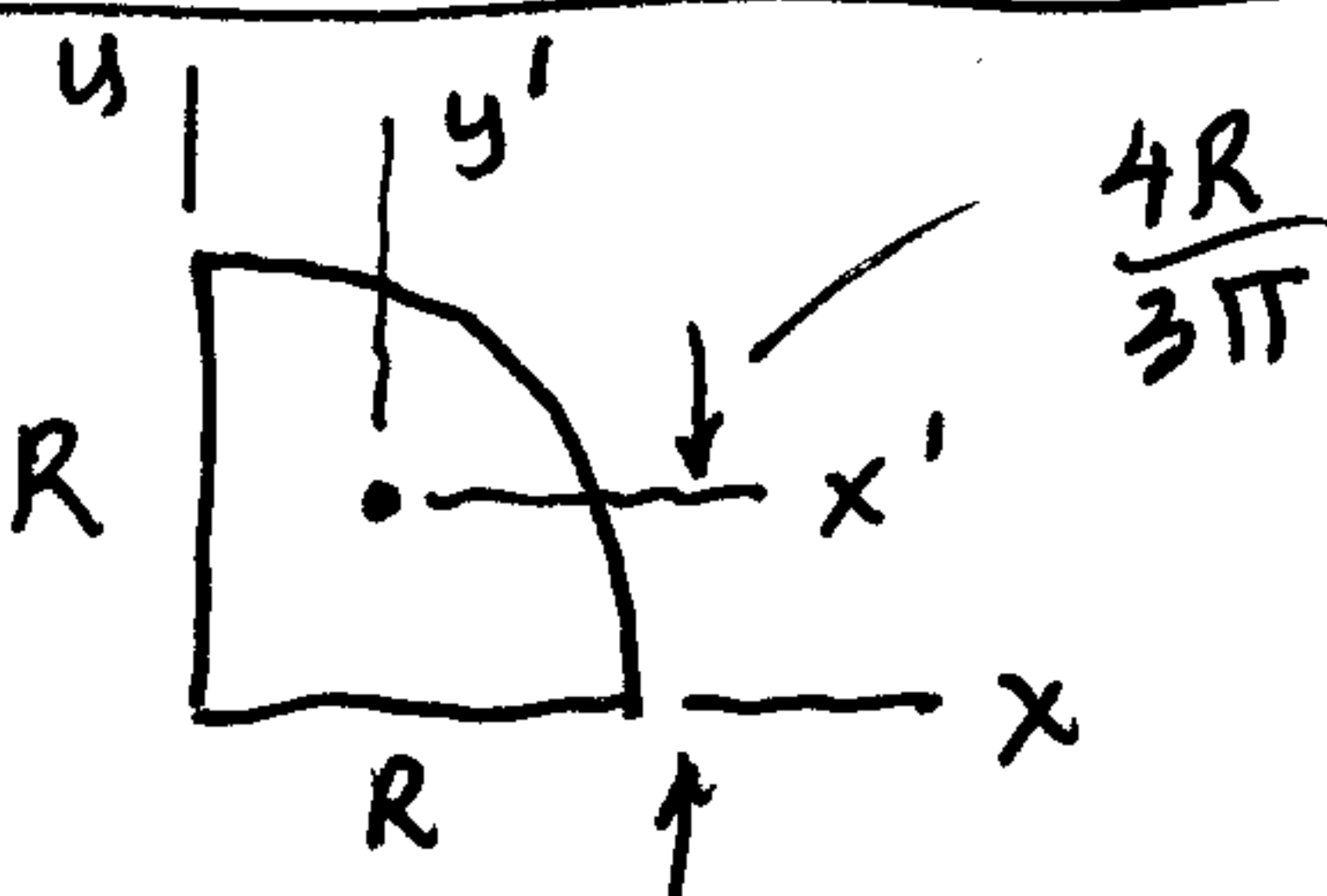
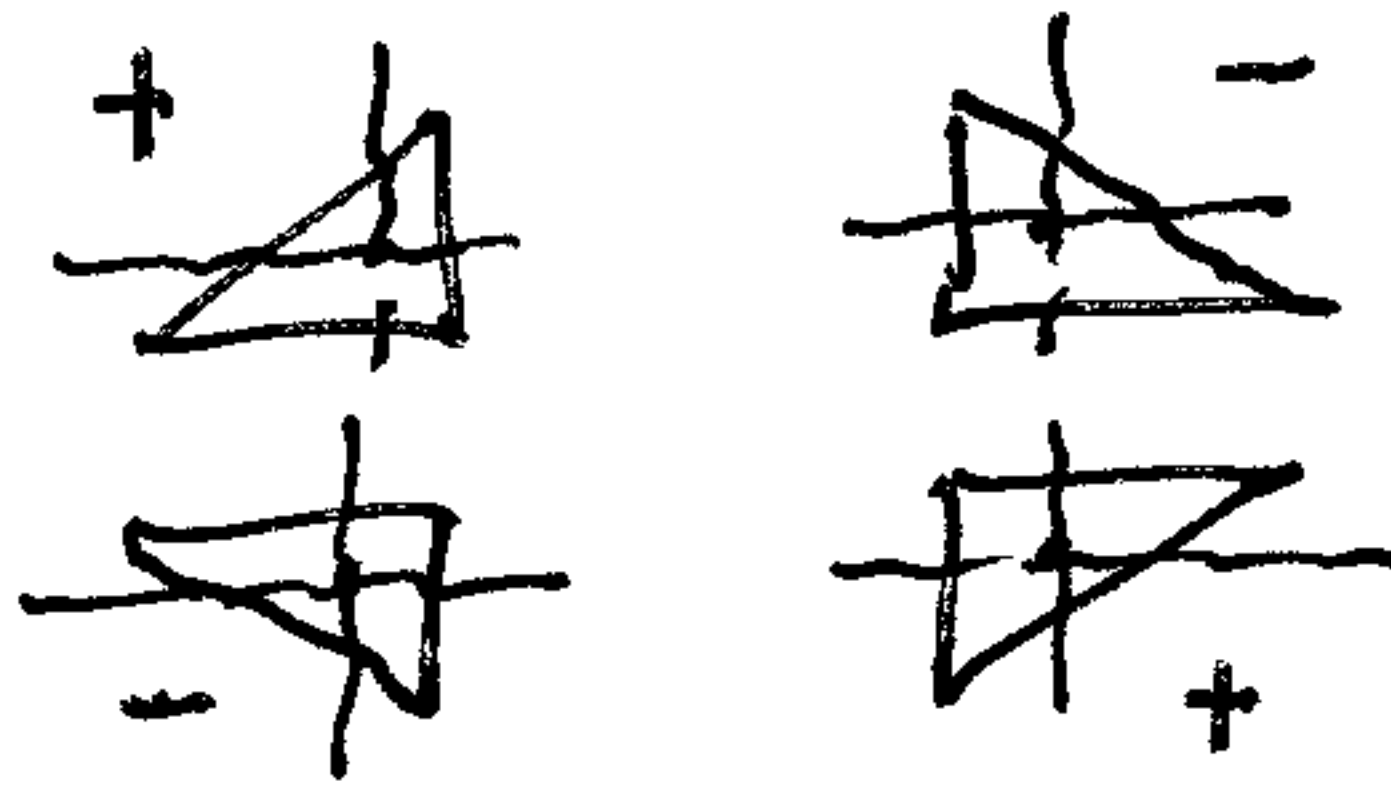
Signs!



Gives

$$\bar{I}_{x'y'} = \frac{1}{72} b^2 h^2$$

But does not give orientation signs:

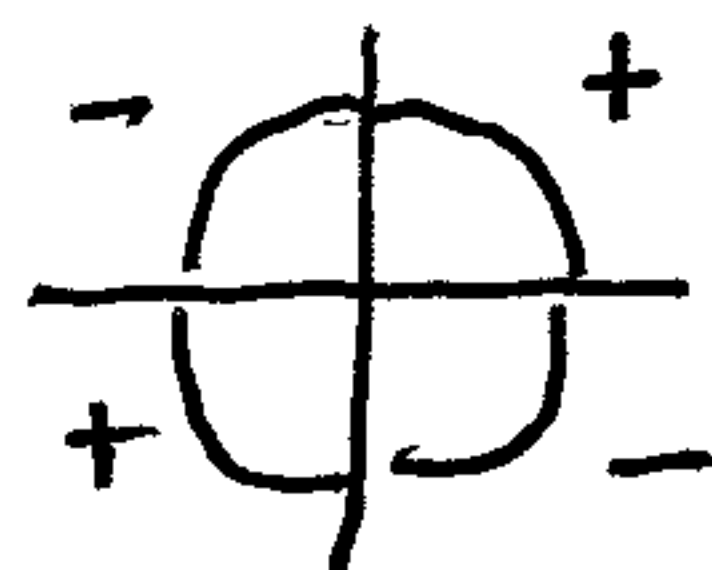


$$I_{base} = \frac{\pi}{16} R^4$$

Gives I_{origin} :

$$I_{xy} = \frac{1}{8} R^4$$

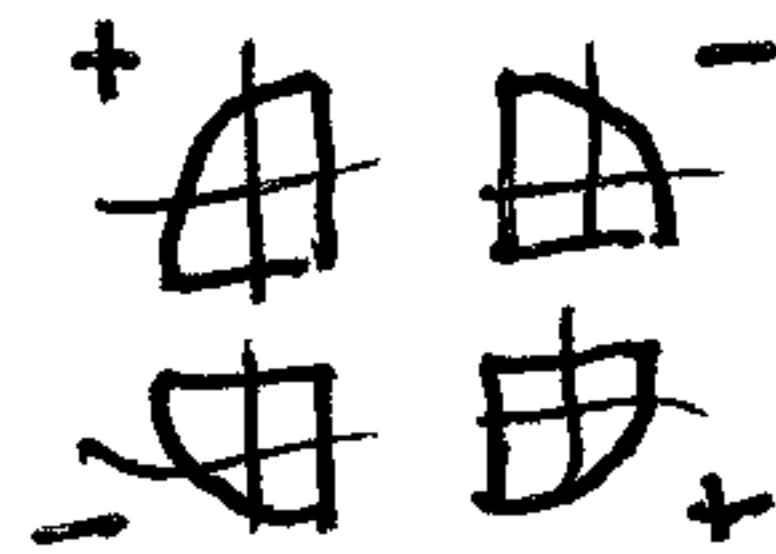
Signs!



Does not give

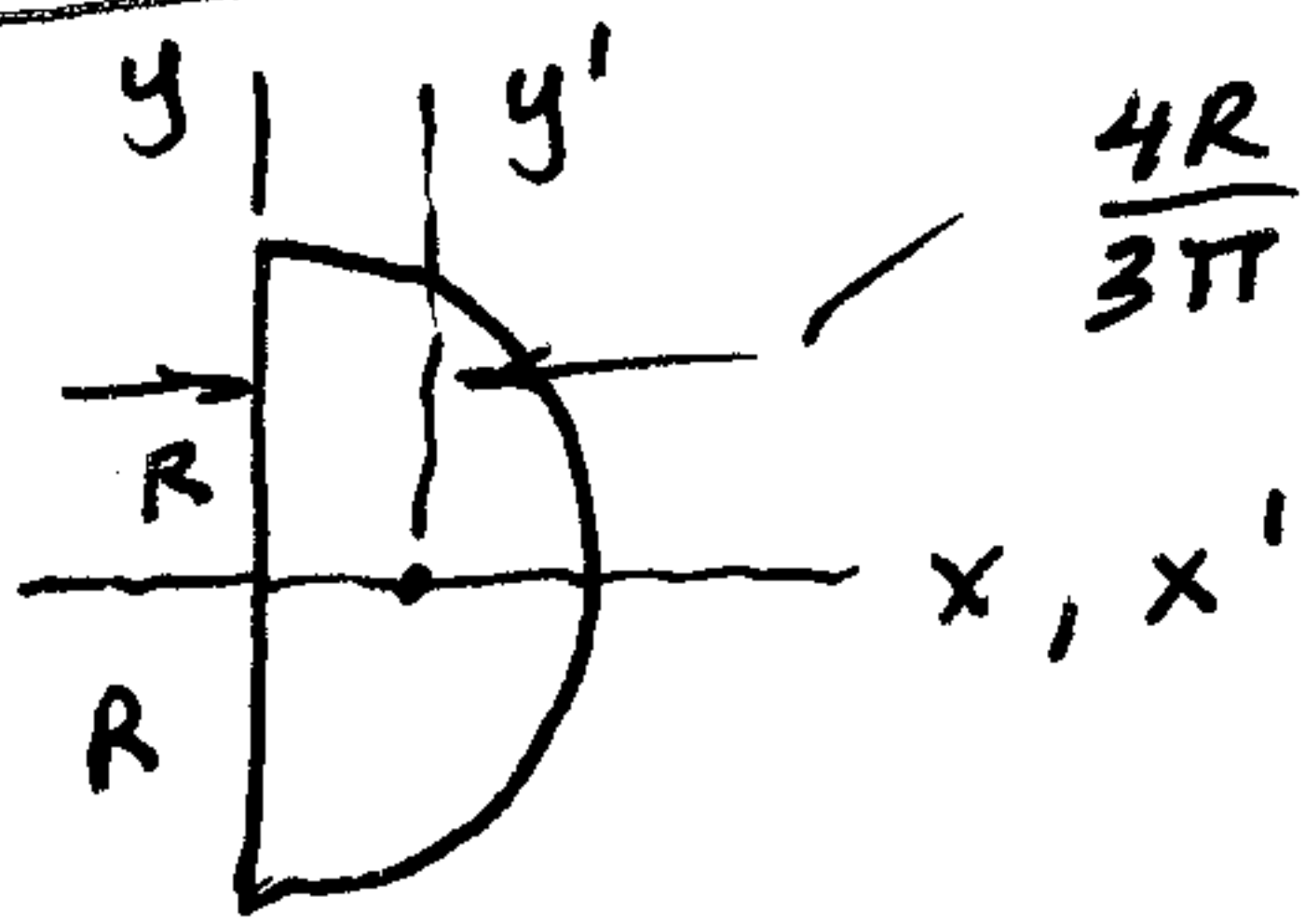
$$\bar{I}_{x'y'} = .01647 R^4$$

Signs!



Crib Sheet Review

(5)



$$I_x = I_{x'} = \frac{\pi}{8} R^4$$

$$I_y = I_{base} = \frac{\pi}{8} R^4$$

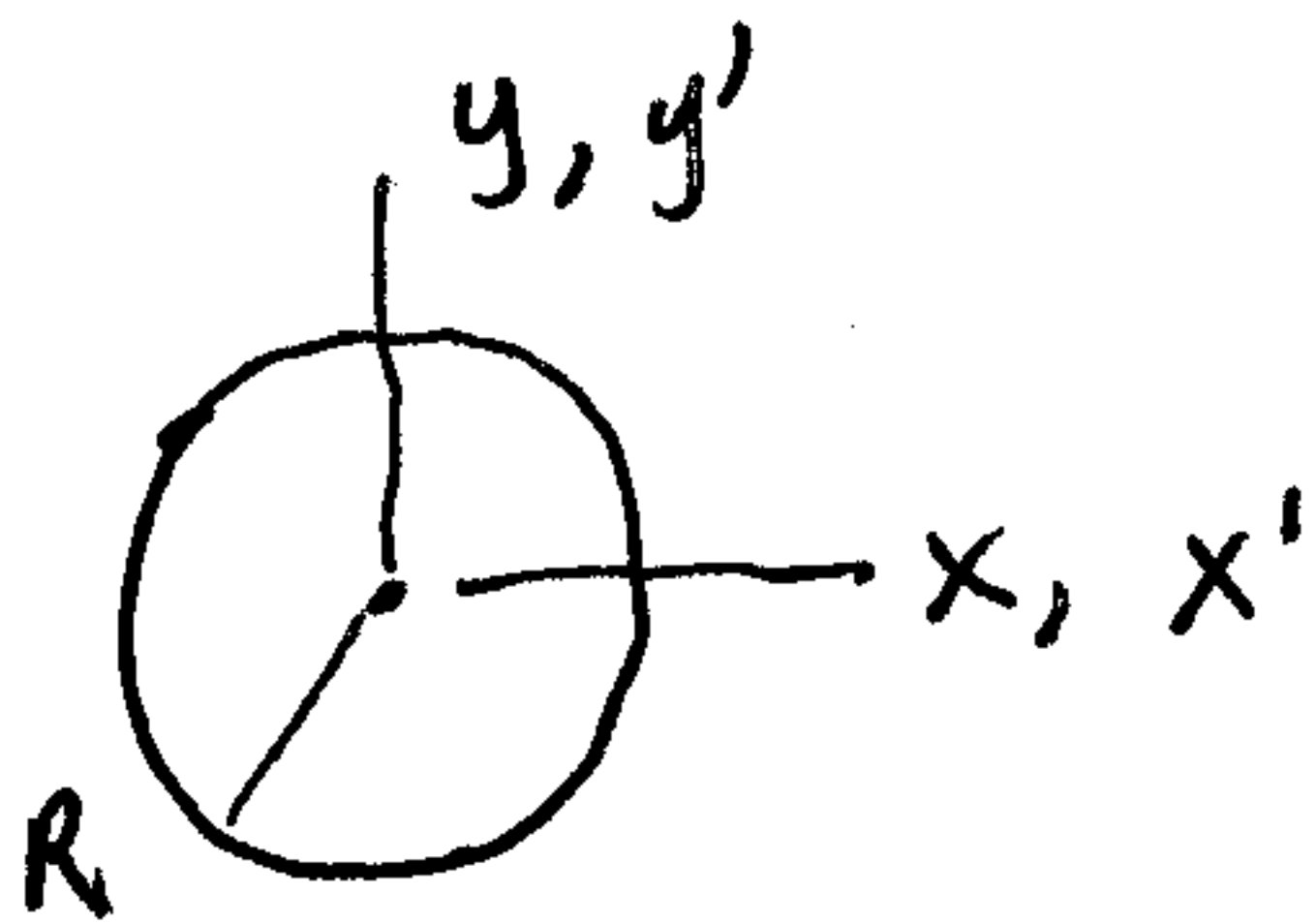
$$I_{y'} = \left(\frac{\pi}{8} - \frac{8}{9\pi} \right) R^4 = .1098 R^4$$

POI:

$$I_{xy} = 0$$

$$\bar{I}_{x'y'} = 0$$

Symmetry



$$\bar{I} = \frac{\pi}{4} R^4$$

$$\bar{I}_{x'y'} = 0$$

Symmetry